

## PANCAKES AND TERNARY STRINGS

MATH 300

**Problem 1:** Lazy Caterer Problem: Let  $c_n$  denote the maximum number of pieces a circular pancake can be cut into with  $n$  cuts. What are  $c_0, c_1, c_2, c_3, c_4$ ? By looking at the differences  $c_n - c_{n-1}$ , make a hypothesis for a recurrence relation for  $c_n$ .

**Problem 2:** How many binary strings of length  $n$  that contain 00 are? Call this  $b_n$ .

- (1) Consider three distinct ways a length  $n$  binary string could start: 1, 01, or 00. For each of these cases, how many ways are there to fill in the rest of the  $n - 1$  or  $n - 2$  positions in the string?
- (2) One of these three cases is not like the others. Which one?
- (3) Use the above answers to write  $b_n$  as a second order linear non-homogenous recurrence relation? What are the initial conditions?